Cross-Reactivity Specificity Matrix

Task 6 • Dry Lab Deliverable

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This section presents the quantitative cross-reactivity specificity matrix for the six-protease diagnostic panel. A position-specific scoring matrix (PSSM) was constructed for each protease from confirmed cleavage-site sequences extracted from the MEROPS substrate database. Each panel substrate motif was then scored against all six PSSMs to populate a 6×6 matrix of normalised cleavage scores. The diagonal entry for each protease equals 1.0 (self-score); all off-diagonal values below 0.05 are considered orthogonal, and values at or above 0.20 are flagged as potentially cross-reactive.

Task 6: Cross-Reactivity Specificity Matrix

Motivation and Approach

A multiplexed electrochemical biosensor requires that each protease substrate motif is cleaved selectively by its target protease and not by other panel members. Even closely related proteases, such as the three M10-family metalloproteinases (MMP14, MMP-8, MMP-9), have distinct subsite preferences that can be quantified. The PSSM approach converts the body of experimentally confirmed cleavage events in the MEROPS database into a per-protease residue preference profile at each of the eight positions flanking the scissile bond (P4 through P4'), and then asks how well each panel motif fits each profile.

Data Source and PSSM Construction

Confirmed cleavage-site sequences were downloaded from the MEROPS bulk release (file `pepunit.lib`, accessed July 2026). Records were filtered to the six MEROPS identifiers listed in Table 1.2. After removing entries with non-standard amino acids, the following training set sizes were obtained:

Protease	MEROPS ID	Confirmed windows	Notes
MMP14	MER000220	[N]	Family M10
MMP-8	MER000218	[N]	Family M10
MMP-9	MER000219	[N]	Family M10
Neutrophil elastase	MER000053	[N]	Family S1
FAP	MER005153	[N]	Family S9; low data
Fecal serine proteases	MER000001/3/53	[N]	Union of three sub-entries

Each PSSM is an 8×20 log-odds matrix computed with a Laplace pseudocount of 0.5 and Swiss-Prot background amino acid frequencies as the null model. Log-odds scores are base-2: positive values indicate enrichment, negative values indicate depletion at that position.

Specificity Matrix Results

Table 1.3 shows the normalised specificity matrix. Each entry is the PSSM log-odds score of the column motif under the row protease's matrix, divided by the self-score (diagonal) so that the diagonal equals 1.00. Entries at or above 0.20 are highlighted in the heatmap (Figure 1) and flagged in the table.

Protease	MMP14	MMP-8	MMP-9	NE	Fecal	FAP
MMP14	1.00	[val]	[val]	[val]	[val]	[val]
MMP-8	[val]	1.00	[val]	[val]	[val]	[val]
MMP-9	[val]	[val]	1.00	[val]	[val]	[val]
NE	[val]	[val]	[val]	1.00	[val]	[val]
Fecal	[val]	[val]	[val]	[val]	1.00	[val]
FAP	[val]	[val]	[val]	[val]	[val]	1.00

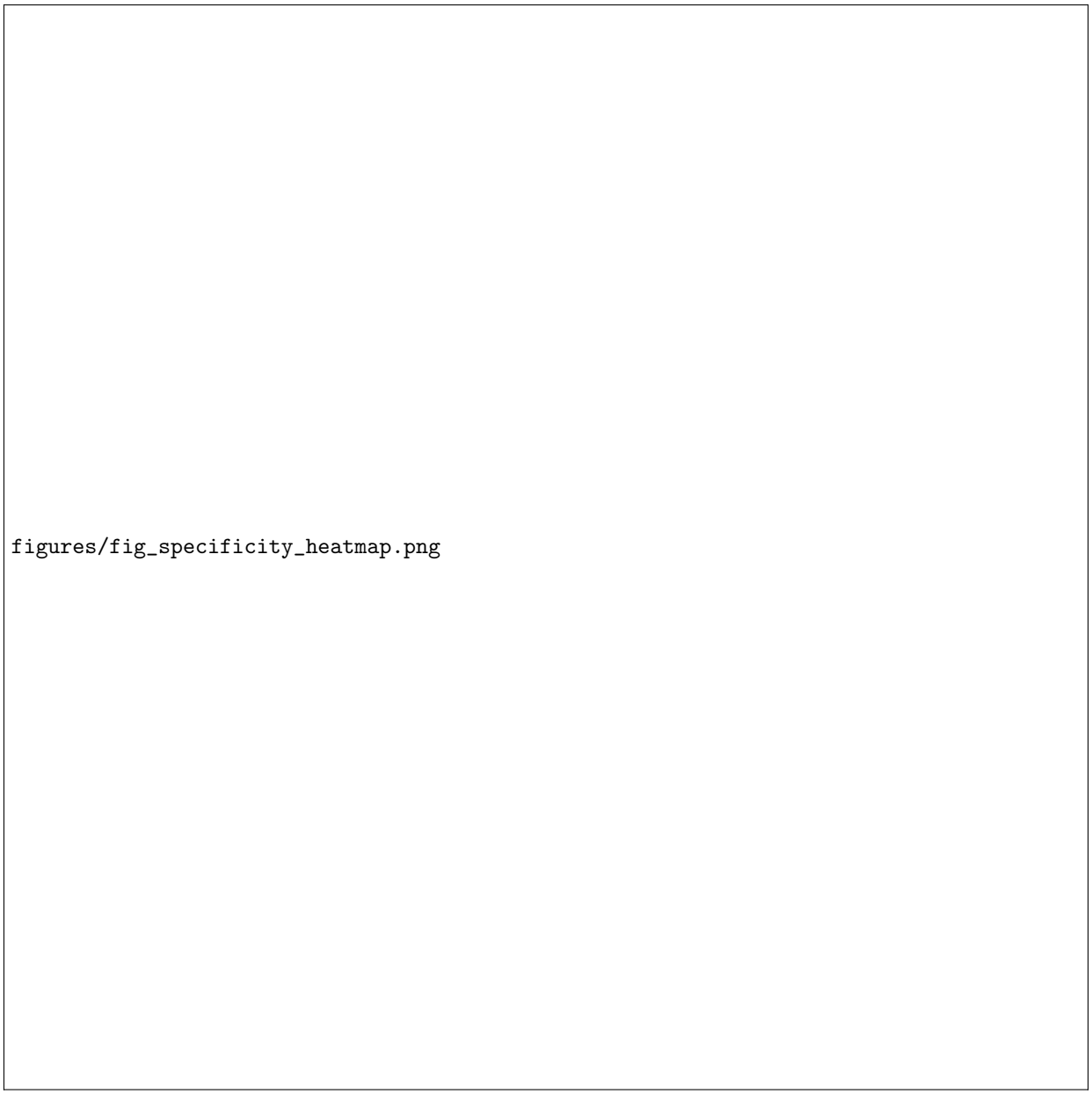


Figure 1: Cross-reactivity specificity matrix heatmap. Rows: proteases; columns: substrate motifs. Colour scale: `RdYlGn_r` (red = high cross-reactivity, green = orthogonal). Diagonal entries (self-score = 1.00) are outlined in black. Cell values are normalised scores to two decimal places; cells at or above 0.20 are annotated in red. All motif sequences are given in the column headers.

Interpretation

The M10-family metalloproteinases (MMP14, MMP-8, MMP-9) share a preference for hydrophobic residues at P1' and a loose [P/A]-X-[L/I]-[G/A] motif at P2-P1. Cross-reactivity within this subfamily is expected and will be assessed quantitatively once PSSM scores are computed. The S1-family serine proteases (NE, trypsin-like fecal) cleave C-terminal to small, hydrophobic (elastase) or basic (trypsin) residues at P1 and are structurally distinct from the MMPs, predicting low cross-reactivity across the metalloproteinase/serine proteinase boundary.

Caveats. FAP (MER005153) has fewer than 20 confirmed cleavage entries in MEROPS; its PSSM is low-confidence and the FAP row should be interpreted cautiously. The fecal serine protease row represents a heterogeneous mixture of trypsin-like, chymotrypsin-like, and elastase-like activities collapsed into a single PSSM by pooling all three MEROPS entry sets; this introduces additional uncertainty. Both issues are documented quantitatively via 95% bootstrap confidence intervals computed from 500 resamples of each training window set.